

Open Source



Teaching Objectives:

Understand the basics of Information and Communication Technologies (ICT).

Discover the ecosystem of Free and Open Source Software (FOSS).



summary

Chapter 1: Information Technologies

1.1 Definitions

1.2. Tools

1.2.1. Hardware

1.2.2. Software

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Chapter 2: Open Source Tools

2.1 Introduction

2.2 Development environment

2.3 Office software (LibreOffice Suite)

2.4 Collaboration (Storage and sharing)

2.5 Contributing to an open source project



Chapter 1: Information Technologies

1.1 Definitions:

1. Computer Science:

The study of how information can be automatically processed, stored, and transmitted using algorithms, computers, and software. We can say that computer science is a discipline that solves real-life problems through automation.

2. Information Technology (IT):

The use of hardware and software tools to manage data and support organizational processes.

3. Information and Communication Technologies (ICT):

A broad term that includes IT and communication systems (Internet, mobile networks, wireless communication) that enable information exchange.



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1.2 Tools:

A computer system is made up of two entities: hardware and software.

a. Hardware: this is the physical part of a computer, it includes the hardware components with which a computer is built.

b. Software: this is the logical part of a computer, it is made up of all the programs that make it possible to operate the hardware.

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Hardware and software are absolutely complementary and inseparable in a computer system.

1.2.1. Hardware

A computer system can be very simple, for example the only requires a management of a small business often

microcomputer, but it can also become very complex. Large scientific computing laboratories, industries, banks, hospitals, etc. need a large number of powerful computers linked together by networks.

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A computer is normally capable of acquiring and storing information, carrying out processing and restoring stored information.

** Types of computers*

Computers can generally be classified by size, power and areas of application as follows:

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a. **Supercomputers:** Very large and very powerful computers (computing speed) which use tens of thousands of microprocessors to carry out certain tasks such as calculations for weather forecasting, for complex calculations (partly for scientists), to analyze the human genome as well as several other functions.



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b. Embedded computers

Is a component incorporated into a system (car, plane, washing machine, refrigerator, camera, etc.). These computers are built to perform a limited number of tasks (such as guidance or security).



Chapter 1: Preamble – Definitions and Generalities

c. Microcomputers

Represents the most common type of computer on the market. Their power continues to increase. There are different types of microcomputers: desktop computers, portable computers (notepads), pocket computers (PDA, personal digital assistant), Tablet PC and body computers.

Chapter 1: Preamble – Definitions and Generalities

d. Desktop computers

Desktop computers are designed to be used on a desk or table. They are generally larger and more powerful than other types of personal computers. They are made up of separate components. The main component, called a system unit, is usually a rectangular box that sits on or under a desk.

Other components such as the monitor, mouse, and keyboard connect to the system unit.



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e. Laptop

Laptops are portable, lightweight desktop computers with a thin screen. They are often called notebook computers because of their small dimensions. They can run on battery power, which allows them to be taken anywhere. Unlike desktop computers, laptops combine the CPU, screen, and keyboard into a single package. The screen folds into the keyboard when the computer is not in use.



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f. Handheld computers

Pocket computers, also called personal digital assistants (PDAs), are battery-powered computers that, due to their small size, can be taken almost anywhere. Although not as powerful as desktops or laptops, handhelds are convenient for scheduling appointments, storing addresses and phone numbers, and playing games.

Some offer advanced features, for example allowing you to make phone calls or access the Internet. Instead of a keyboard, handheld computers have a touch screen that you can operate with your finger, a stylus (a pen-shaped pointing tool), or a digital pen.



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g. Tablet PC

The Tablet PC is a laptop computer like a large cell phone with a touch screen. It usually uses an on-screen virtual keyboard, stylus or digital pen. Typically, these tablets do not need a keyboard. There are two popular categories of tablet PCs:

- Slate Tablet PC: is a type of tablet where the keyboard is not connected.
- Convertible tablet: is basically a laptop with a screen that can rotate and fold over the keyboard to create the tablet.



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h. Body computers

These computers are small and light because they must be carried on the human body. Body computers come in the form of bracelets, pendants, glasses and rings. Wearing such devices allows constant interaction with the computer system. Additionally, these devices are multi-tasking. You can perform other routine tasks at the same time.

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This computer can be used as a small chip programmed to monitor the movements of animals. A pre-programmed lightweight microprocessor chip is stapled to the animal's body, like ears. Such a chip monitors the movement of animals in a given region.



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Communication Networks:

Router: connects networks and directs Internet traffic.

Switch: interconnects devices inside a local network.

Modem: connects a local network to the Internet.

Fiber optics: provides high-speed Internet using light.

Wi-Fi: wireless connectivity for mobile devices.

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Peripherals:

There are different types of devices that are generally classified into 03 types:

- Input devices
- Output devices
- Input-output devices

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I. Input Devices:

The input devices are used to provide information (or data) to the computer system.

We mainly find there:

1. Keyboard: allows you to write characters (letters, numbers, symbols, etc.). It is therefore an essential input device for the computer, because it is thanks to it that we can send commands.

2. Mouse: allows you to move the pointing cursor on the screen and allows you to select, move, and manipulate objects using buttons.

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3. Scan: The scanner is a device that allows you to scan documents from a "paper" format and generate documents in "electronic or digital" format that can be saved on your computer. The scanner is characterized by its scanning quality (resolution).

4. Webcam: is a camera designed to be used as a computer peripheral, and which produces a video whose purpose is not to achieve high quality, but to be able to be transmitted live through a network, typically the Internet.

5. Microphone: A microphone (or simply mic) is a system for converting sound waves into electrical impulses. The electrical signal has the advantage of being easily processed (see sound reinforcement and sound recording).

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6. Optical disc drive: It allows you to read digital data (computer data). It can be a CD-ROM player, a DVD-ROM player, a Blue Ray player, etc.

II. Output devices

Output devices are used to output information from the computer system, they mainly include:

1. Screen: Allows you to view information coming from the computer.

2. Printer: A printer transfers data from a computer to a medium (paper). A printer is not needed to use the computer, but it is very useful for printing e-mail, letters, invitations, announcements, etc.).

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3. Speakers: Speakers are used to reproduce and increase the volume of sound. They can be built into the case of the system unit or connected using cables. Speakers allow you to listen to music and hear sound effects from the computer.

III. Input-output devices

We can also encounter input-output devices that operate both in writing and reading: a hard disk, an optical disk reader/writer or a USB key, for example, can store data (output) as well as read it (input).

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1. USB key: A USB flash drive is a removable storage medium that plugs into a computer's USB (Universal Serial Bus) port. A USB flash drive contains flash memory and has little or no mechanical components, making it highly shock-resistant. It does not require a battery as it is powered by the USB connection.

2. External hard drive: is a hard drive embedded in a solid, easily transportable case that connects to a computer via USB.

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3. Optical Disc Burner Reader: It allows you to read or write digital data (computer data). It can be a CD-ROM burner reader, a DVD-ROM burner reader, a Blue Ray burner reader, etc.

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Smart Chips

RFID (Radio Frequency Identification): used in student ID cards for library access.

NFC (Near Field Communication): used in contactless payments.

Embedded processors: control cars, drones, or smart appliances.

IoT sensors: measure environmental data (temperature, air quality).

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1.2.2 Software

1. Operating Systems (OS):

The operating system is a set of programs that directs the use of resources hardware of a computer by application software (text editor, web browser, media player, video game, ...). The operating system manages the requests as well as the necessary resources. The operating system serves as an intermediary between the software and the computer hardware.

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a. Operating system components

The operating system is composed of a set of software that manages interactions with the hardware. Among this set of software, we generally distinguish the following elements:

- **The core (kernel):** representing the fundamental functions of the operating system such as memory management, processes, files, major input-output, and communication features.
- **The command interpreter (shell):** allowing communication with the operating system through a command language. Its function is to interpret the commands that a user types on the keyboard in the command line interface.
- **The file system :** allowing files to be saved in a tree structure.

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a. Operating system components

- **The user interface** : allows a user to operate the machine.
- **The drivers** : allowing the operating system to interact with devices.

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b. Types of operating systems:

The four main types of operating systems are :

Single-User, Single-Tasking Operating Systems : These operating systems allow only one user to work on the computer at a time, and they can handle only one task or program at a time. Examples include early versions of MS-DOS.

Single-User, Multi-Tasking Operating Systems : These operating systems allow one user to run multiple programs or tasks simultaneously. The operating system switches between tasks quickly, giving the illusion of running multiple programs at once. Examples include modern versions of Windows, macOS, and Linux.

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b. Types of operating systems:

Multi-User Operating Systems : These operating systems allow multiple users to work on the same computer simultaneously. Each user can have their own user account and run their own programs independently. Examples include UNIX-based systems like Linux and macOS, as well as server versions of Windows.

Real-Time Operating Systems (RTOS) : These operating systems are designed to process data and respond to events in real-time, with strict deadlines for completing tasks. They are used in applications where timing is critical, such as controlling industrial machinery, automotive systems, and aerospace applications. Examples include QNX and VxWorks.

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There are plenty of Operating Systems available in the market which include paid and unpaid (Open Source). Following are the examples of the few most popular Operating Systems:

Windows: This is one of the most popular and commercial operating systems developed and marketed by Microsoft. It has different versions in the market like Windows 8, Windows 10 etc and most of them are paid.

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Linux This is a Unix based and the most loved operating system first released on September 17, 1991 by Linus Torvalds. Today, it has 30+ variants available like Fedora, OpenSUSE, CentOS, UBuntu etc. Most of them are available free of charges though you can have their enterprise versions by paying a nominal license fee.

MacOS This is again a kind of Unix operating system developed and marketed by Apple Inc. since 2001.

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iOS This is a mobile operating system created and developed by Apple Inc. exclusively for its mobile devices like iPhone and iPad etc.

Android This is a mobile Operating System based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets.

Some other old but popular Operating Systems include Solaris, VMS, OS/400, AIX, z/OS, etc.

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2. Business Application Software:

Given the ongoing technological revolution, most manual and labor-intensive logistics and workforce operations have now transitioned to digital platforms.

This is where **Business Application Software (BAS)** plays a crucial role. From time tracking to customer relationship management, such software simplifies every task while ensuring that all relevant details are systematically documented.

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Different Types of Business Application Software:

Enterprise resource planning (ERP) software :

Enterprise Resource Planning (ERP) is a software system that helps organizations automate, integrate, and manage their core business processes such as accounting, production, supply chain, and human resources through a single, unified platform. It ensures data consistency and reliability, improves communication between departments, and enhances overall operational efficiency.

Examples:

Oracle NetSuite

Microsoft Dynamics 365

Epicor

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Customer relationship management (CRM) software :

CRM software is the software of choice for a business prioritizing customer interactions and relationships. Driving business growth through data-driven solutions, this tool analyzes the ways your business interacts with your customers identifying the loopholes and filling the gaps.

Examples:

HubSpot CRM

Zoho CRM

Insightly

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Accounting Software :

Accounting software is designed to help organizations manage their financial operations efficiently. It enables users to monitor cash flow and expenses, track tax filings, and ensure the accuracy of financial records and inventory data. Serving as an all-in-one solution, accounting software simplifies financial management, reduces human error, and supports informed business decisions.

Examples:

NetSuite

Zoho Books

Fresh Books

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Business Intelligence (BI) Software

Gain a competitive edge over your rivals with the integration of business intelligence software into your operations. The tool is helpful in increasing the efficiency of your business, visualizing trends, and identifying growth opportunities for rapid business scaling.

Examples:

Zoho Analytics

Microsoft Power BI

Qlik Sense

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Marketing Automation Software

Per the name, marketing automation software is the one that keeps a keen eye on your business's marketing campaigns. Ensuring a structured flow, this tool is best used for scheduling your marketing posts and emails, catering to your business's marketing needs at every level.

Examples:

Mail Chimp

HubSpot

Hootsuite

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Resource Management Software

Resource management software is another tool that efficiently tracks employee performance and monitors the company's financial accounts and overall performance. The tool best serves the company's managerial authority in assigning and allocating tasks and equipment to the specific resource,

Examples:

monday.com

Float

Runn

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Payment Transaction Software

Automate your payment transaction for better-structured workflows with the payment transaction software. The tool helps maintain a digital record of your resources' payments, pending or released, bringing efficiency throughout. What's more, the tool even helps you keep track of any pending invoices or payments from your customers, ensuring accuracy in the business's finances.

Examples:

Stripe

PayPal

Go Cardless

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3. Office software:

1. Microsoft office:

Microsoft Office is one of the most popular office suites in the world. Over the years, it has evolved to meet diverse needs, both for individuals and businesses.

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*** The classic version: Microsoft Office:** Traditionally, Microsoft Office was offered as perpetual licenses. This means that users purchased a specific version of the office suite, which they could use indefinitely without any subsequent updates included. This classic version includes popular applications such as:

Word: is a word processing software and used to create, edit, format, and print text documents. It is one of the most popular programs for writing letters, reports, resumes, articles, and many other types of documents.

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Excel : is a spreadsheet that is part of the Microsoft Office suite. This tool allows the creation of worksheets where complex calculations, data analysis, and charts can be performed. Widely used in businesses, it facilitates the management of large amounts of data thanks to its advanced functions. Its integration with other Microsoft applications, such as Word and PowerPoint, enhances productivity.

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PowerPoint: is part of the Microsoft Office suite. PowerPoint allows users to create presentations in the form of slides, usually displayed through a video projector to support an oral presentation. It is possible to integrate text, images, animations, tables, and charts. It is the replacement for overhead projectors and transparencies.

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OneNote: is a digital notebook that allows you to capture and organize information across all your devices. Jot down your ideas, keep track of class and meeting notes, web clippings, or make a to-do list, as well as draw and sketch your ideas.

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Outlook: is a software from the Microsoft Office suite that allows the receiving and sending of emails, organizing them, as well as managing a complete address book, handling tasks, and managing your time. In a business setting, Outlook can be connected so that you can check others' availability to schedule an appointment.

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2. Collaboration tools: Collaboration in business increasingly relies on efficient digital tools. Microsoft 365 and Google Workspace are the two most popular solutions. These two platforms offer integrated productivity, collaboration, and communication tools. They include word processing software, spreadsheets, storage solutions, and communication applications.

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*** Microsoft 365:**

With the rise of collaborative work and online tools, Microsoft introduced Microsoft 365, a subscription-based solution that has revolutionized the Office landscape. With Microsoft 365, users have access to online applications, collaboration tools such as Teams and SharePoint, as well as regular updates. This version is particularly advantageous for businesses, as it: Improves productivity Facilitates communication within teams.

- Outlook (emails and calendar)
- Word, Excel, PowerPoint (office applications)
- Teams (video conferencing and collaboration)
- OneDrive and SharePoint (file storage and sharing)

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*** Google Workspace:**

Google Workspace is the professional office suite developed by Google, specifically designed to meet the growing needs of businesses for productivity and collaboration. Available exclusively as an online service, Google Workspace brings together key applications such as:

- Gmail (emails and calendar)
- Docs, Sheets, Slides (online office tools)
- Google Meet (video conferencing)
- Google Drive (file storage and sharing)

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Unlike many traditional software programs, Google Workspace does not require downloading or local installation on your computer. You simply need an Internet connection and a Google Workspace account to access all the tools from any compatible web browser.

For mobile users, dedicated mobile applications are available for Android and iOS, allowing access to Google Workspace from a smartphone or tablet.

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1.3. Application:

Information technologies support a wide range of digital applications that improve communication, document sharing, business workflows, and organizational productivity. Below is an extended classification of four major application domains.

1.3.1. Communication Spaces:

Communication networks enable users to exchange data and access digital resources. Three main types are utilized in modern organizations:

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- **Internet – *Public Global Network:***

The Internet is a worldwide network that interconnects millions of computers and devices, providing universal access to information, communication services, digital platforms, and online resources.

- Open and publicly accessible
- Borderless and global connectivity
- Supports web browsing, cloud computing, e-commerce, e-learning, etc.

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- **Intranet – Internal Private Organizational Network**

An intranet is a secure, private network operated within an institution or company. Its purpose is to share internal resources, streamline workflow, and ensure controlled communication among members.

- Internal document distribution
- Private communication channels
- Access to academic and administrative services

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- **Extranet – Extended Restricted Network for Authorized External Users**

An extranet is a portion of the intranet that is accessible to selected external stakeholders such as partners, suppliers, parents, or collaborating institutions.

- Enhances inter-institutional col
- Shares specific data securely with external entities

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1.3.2. Multimedia Communication Technologies:

Multimedia systems combine text, image, sound, and video to support interactive communication and remote collaboration.

- **Videoconferencing:**

A technology that enables real-time video communication between distant participants using the Internet.

Example Tools:

Zoom, Microsoft Teams, Google Meet

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- **Audioconferencing**

A communication method based on voice-only connections without video transmission, using telecommunication networks or VoIP(**Voice over Internet Protocol**)

Example Tools:

Skype,WhatsApp calls, Zoom audio calls, Google Voice, Viber.

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1.3.3. Electronic Data Interchange (EDI):

EDI is a technology that enables the automated exchange of structured business documents between organizations' information systems without paper-based processing.

- **Advantage:**

Speed, Accuracy, Cost reduction, Reliability.

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1.3.4. Workflows and Business Process Automation:

A workflow is a defined sequence of automated tasks performed according to organizational rules to ensure that the correct action is conducted at the right time.

- **Advantage:**

- Eliminates paperwork

- Prevents approval delays

- Tracks each step with timestamps

- Improves organizational productivity